

Protein Mass Calculator from CS 171 Homework 1

Honor Code: Solve this assignment individually; do not collaborate with classmates or look for online solutions. We check for code plagiarism. The assignment is governed by the College HonorCode and Departmental Policy. Remember, any code you submit must be your own; otherwise, you risk being investigated by the Honor Council and facing the consequences of that. Finally, please remember to have the following comment included at the top of each of your .java file:

```
/* THIS CODE WAS MY OWN WORK, IT WAS WRITTEN WITHOUT CONSULTING  
CODE WRITTEN BY OTHER STUDENTS OR COPIED FROM ONLINE RESOURCES.  
_Your_Name_Here_ */
```

1.  Protein Mass Calculator. Each amino acid has a molecular mass. Given a protein sequence represented as a string of single-letter amino acid codes (like "ACDE"), write a Java program ProteinMassCalculatorFull(String protein) to calculate its total molecular mass. The amino acid mass table is provided below.

 Full Amino Acid Mass Table (Average)

Code	Name	Mass
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A	Alanine	89
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R	Arginine	174
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N	Asparagine	132
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D	Aspartic Acid	133
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C	Cysteine	121
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Q	Glutamine	146
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E	Glutamic Acid	147
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G	Glycine	75
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H	Histidine	155
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I	Isoleucine	131
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L	Leucine	131
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K	Lysine	146
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M	Methionine	149
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F	Phenylalanine	165
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P Proline 115

S Serine 105

Code Name Mass

T Threonine 119

W Tryptophan 204

Y Tyrosine 181

V Valine 117

Note your program should return a message “Invalid amino acid” if the input contains invalid amino acid codes.